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## **Assessment of Thermal Maturity Using Palynomorph Darkness Index (PDI) in the Early Paleozoic of Saudi Arabia**

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### **Abstract**

Palynomorph Darkness Index (PDI) is a fully quantitative optical method for assessing thermal maturity which has been established as a rapid and low-cost ‘add-on’ to palynostratigraphic investigations, especially in pre-Devonian sediments. Here, we applied the PDI method to Silurian strata in Saudi Arabia to enhance calibration of Acritarch PDI to vitrinite reflectance ( $VR_o$ ) via Vitrinite Reflectance Equivalents (VREs) such as pyrolysis-based  $T_{max}$  determination, and graptolite and chitinozoan reflectance. Unoxidised palynological concentrates were obtained from core samples from various sections in Saudi Arabia from the early Silurian (Qalibah Formation) and a maturity range from immature through the oil, condensate and gas windows to the postmature stage. PDI measurements were taken on suitable acritarch taxa using a transmitted light optical microscope and a digital camera. The PDI values obtained were correlated with VREs derived from  $T_{max}$ , graptolite reflectance and chitinozoan reflectance. Published PDI –  $VR_o$  correlations established in Devonian-Carboniferous sections in the eastern USA were also considered. The present results confirm that acritarch PDI can be successfully applied for thermal maturity estimation, especially in pre-Devonian age, organic-rich sediments, where most of the available alternative methods show weak correlation to  $VR_o$ . The most suitable acritarchs for PDI determination are the *Veryhachium* / *Neoverhachium* Group. Preliminary results suggest that four distinct stages can be recognized in the  $VR_o$ /VRE: PDI correlation trendline. Data gathered so far allowed a preliminary correlation trendline to be established between PDI and  $VR_o$ /VRE values which can be used for determining thermal maturity ranges based on simple quantitative measurements of the degree of palynomorph darkness under optical microscopy during routine palynological analyses, though further work is needed to refine PDI vs.  $VR_o$ /VRE in rocks in the upper range of thermal maturity.

### **Introduction**

Vitrinite Reflectance ( $VR_o$ ) is the standard method used to estimate thermal maturity in oil and gas exploration but this technique cannot be applied to pre-Late Devonian rocks, none of which contain vitrinite. Various proxies for  $VR_o$  (‘Vitrinite Reflectance Equivalents’ abbreviated to VREs) have been developed, including methods based on pyrolysis-derived  $T_{max}$  and graptolite reflectance.  $T_{max}$ -based

VREs have been widely applied to early Paleozoic rock sections in Saudi Arabia but a comprehensive review by [Evenick \(2021\)](#) has clearly demonstrated that, although these show a strong correlation to  $VR_o$  in the oil and condensate windows, correlation is very poor outside this maturity range. The relatively few graptolite reflectance-based VRE determinations ( $VRE_G$ ) from Saudi samples appear to correlate to  $VR_o$  more consistently but may also be unreliable within the dry gas to postmature maturity range due to problems associated with graptolite bireflectance.

This paper describes results from a project which evaluated the potential of Palynomorph Darkness Index (PDI), another proxy for  $VR_o$ , in terms of its ability to produce rapid and inexpensive estimates of thermal maturity of early Paleozoic sections, preliminary results of which were published by [Clayton et al. \(2024\)](#). Palynomorphs (spores and pollen, and fossil phytoplankton such as acritarchs) gradually change color and darken when heated, eventually becoming black. As this darkening is irreversible, the color of palynomorphs has long been utilized to assess thermal maturity. However, characterization of color is complex and color cannot easily be described by a single parameter. This has resulted in many schemes being established which attempt to correlate palynomorph color to maturity but none of these are quantitative. To overcome this difficulty, Palynomorph darkness Index (PDI) was introduced, a fully quantitative method that converts color to greyscale which can then be expressed on a simple 0-100% scale ([Goodhue and Clayton, 2010](#)). PDI was successfully employed in several small-scale projects, mainly using spores from Upper Paleozoic sections ([Clayton et al., 2017](#); [Spina et al., 2017](#)).

A major breakthrough in refining the calibration of spore/pollen PDI to  $VR_o$  was the publication of a large number of paired VR and spore/pollen PDI results from TotalEnergies by [Buratti et al. \(2024\)](#) which revealed a complex but useful relationship between PDI and VR comprising four distinct stages. A parallel four stage trend to that described for spores and pollen was noted when acritarch PDI is correlated to  $VR_o$  ([Clayton et al., 2024](#)) ([Fig. 1](#)). In the least mature of these, PDI Stage 1 ( $VR_o/VRE$  0.2 – 0.8%), PDI is <10%. In the second stage ( $VR_o/VRE$  0.8 – 0.9%), Acritarch PDI increases very rapidly from ca. 10% – ca. 25%. Stage 3 ( $VR_o/VRE$  0.9 – 2%) shows a gradual increase in PDI from ca. 25% to ca. 80% and in Stage 4 ( $VR_o/VRE$  > 2%), PDI increases only very slowly with increasing maturity, from ca. 80% to ca. 95%. The calibration of Acritarch PDI to  $VR_o$  is poor in stages 3 and 4, relying exclusively on Tmax- based VREs and  $VRE_G$ . Direct correlation of Acritarch to  $VR_o$  is obviously the preferred method but obtaining suitable material to from the same samples is difficult, as vitrinite does not appear in the stratigraphic record until late Devonian time and acritarchs almost became extinct in latest Devonian time, severely limiting the number of sections suitable for study.

A relatively little-known method using chitinozoan reflectance ( $R_{ch}$ ) as a proxy for  $VR_o$  was published by [Tricker et al. \(1992\)](#), who established a strong correlation between  $R_{ch}$  and  $VR_o$  through a wide range of maturity. This technique was applied successfully by [Marshall \(1995\)](#), who published a comprehensive interpretation of the thermal history of Lower Palaeozoic sections in five exploration wells in northern Saudi Arabia. Acritarch PDI results from two of these wells are presented and their significance is discussed in this paper.

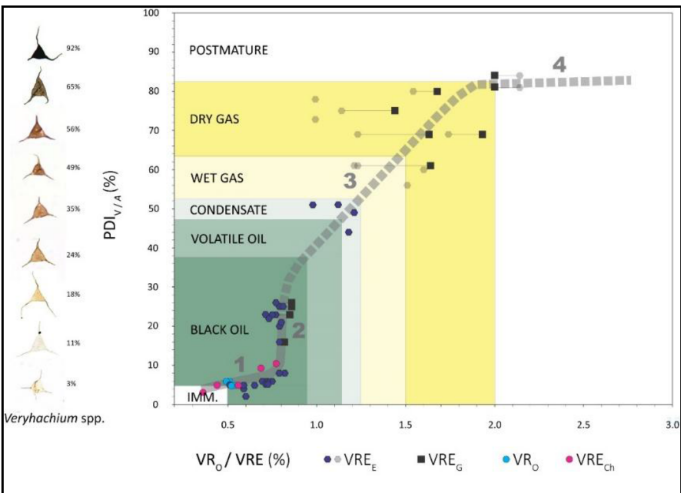


Figure 1—Correlation of acritarch PDI to  $VR_O$  and selected VREs (modified from Clayton, Vecoli, Pan Luo et al., 2024) including new data. A provisional best-fit line is shown, together with PDI stages 1-4. Also included are photomicrographs of typical acritarchs with PDI indicated

Samples and Methodology

Slides of organic residues from six core samples which had been prepared using standard palynological techniques were examined in transmitted light and PDI values were determined using the methodology described by Goodhue and Clayton (2010) and employing the same Nikon microscope and GryphaxArktur digital camera used by Clayton et al. (2025). The six samples investigated were from the same well sections from which chitinozoan reflectance ( $CR_O$ ) had been measured by Marshall (1995) as part of a much larger study. The relative positions of the  $CR_O$  samples published by Marshall, together with our PDI samples are shown in figs 2a and 2b respectively and Table 1.

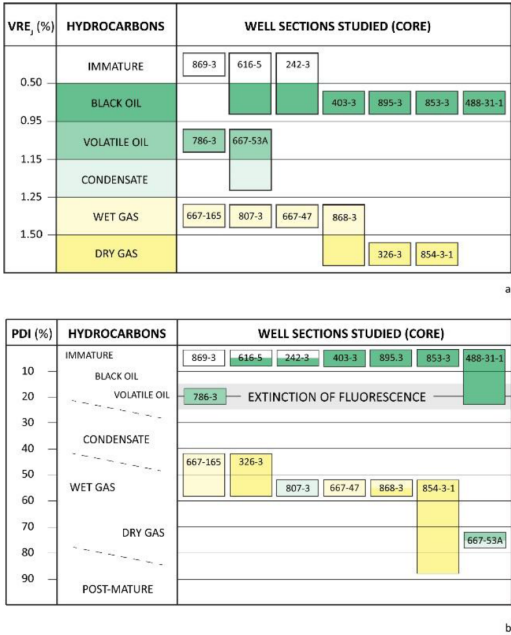


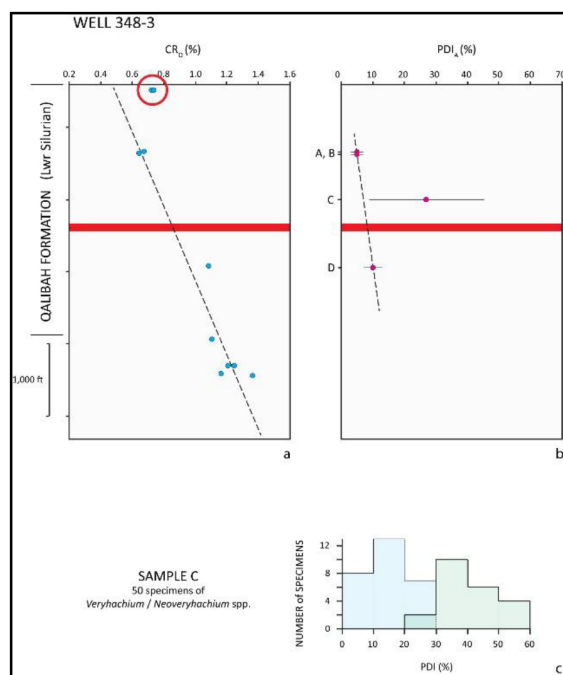
Figure 2—(a) Cored sections investigated for acritarch PDI with maturity assessed mainly by  $T_{max}$ -based VREs and  $VRE_G$  (after Clayton, Vecoli, Pan Luo et al., 2024). (b) Sections retaining same coloring as in Fig. 2 (a), re-ordered on basis of acritarch PDI.

**Table 1—Relative sample positions,  $VRE_{Ch}$  (calculated from  $CR_O$ ), PDI, standard deviation and number of specimens from wells 348-3 and 180-3.**

| WELL 348-3 |                         |                |         |               |                     |
|------------|-------------------------|----------------|---------|---------------|---------------------|
| Sample     | Depth                   | $VRE_{Ch}$ (%) | PDI (%) | Standard dev. | Number of specimens |
| A          | -                       | 0.41           | 5       | 2             | 30                  |
| B          | 25 ft below sample A    | 0.41           | 5       | 2             | 30                  |
| C          | 631 ft below sample A   | 0.46           | 27      | 18            | 50                  |
| D          | 1,571 ft below sample A | 0.63           | 10      | 3             | 30                  |
| WELL 180-3 |                         |                |         |               |                     |
| Sample     | Depth                   | $VRE_{Ch}$ (%) | PDI (%) | Standard dev. | Number of specimens |
| E          | -                       | 0.36           | 3       | 2             | 30                  |
| F          | 915 ft below sample E   | 0.69           | 12      | 5             | 30                  |

## Results

The new Acritarch PDI results obtained are presented in Table 1 and in Fig. 1. Results from Well 348-1 are also shown in Fig. 3(a) and (b). Except for sample C, all the mean PDI values are based on 30 specimens, many of which are from the preferred genera, *Veryhachium* and *Neoverhachium*. These PDI values are fully consistent with previously obtained results (Fig. 1).



**Figure 3—(a) Part of section penetrated by Well 348-3 showing  $CR_O$  (after Marshall, 1995). The anomalous points discussed by Marshall are circled in red. (b) Same section showing acritarch PDI. The red line represents a 110 ft thick quartz dolerite sill. (c) distribution of PDI values from individual acritarchs from sample C.**

The mean PDI obtained from sample C was based on 50 specimens of only *Veryhachium* and *Neoverhachium* spp. This value is considerably higher than expected and is associated with an anomalously high standard deviation. Very few specimens exhibiting a low PDI consistent with the trend in this well were recorded, with most specimens being much darker. The histogram representing the distribution of PDI determinations from individual acritarchs from this sample is clearly bimodal (Fig. 3(c)), suggesting two distinct populations of darker acritarchs.

## Interpretation and applications

**Improved calibration of Acritarch PDI to  $VR_o$ .** Addition of the PDI v.  $VRE_{Ch}$  data from samples A, B, D, E and F to the existing dataset produces a significant improvement to the calibration of Acritarch PDI to  $VR_o$  in PDI Stage 1 (Fig. 1).

**Assessment of relative maturity.** The sections from which the samples investigated by Clayton et al. (2024) are shown in Fig. 2(a) with maturity increasing from top to bottom. These are mainly based on  $T_{max}$  – based VREs, together with a smaller number of  $VRE_G$  determinations. These sections are depicted in Fig. 2(b) with the same colors as in Fig. 1(a) but ordered on the basis of their Acritarch PDI. Many of the wells are ranked in the same order but there are several anomalies, for example Well 667-53A which lies within the Dry Gas Window based on Acritarch PDI but within the Volatile Oil and Condensate windows based on other VREs. Used in this context, Acritarch PDI offers an independent alternative method of estimating maturity.

**Elucidation of thermal history.** The PDI results from samples A, B, D and E are fully consistent with maturation caused by burial. However, the PDI results from sample C are anomalous in terms of both high mean PDI (27%) and high standard deviation (18). Marshall (1995) suggested that anomalously high maturation levels in some samples from the five well sections that he investigated could be attributed deep weathering during the interval of ‘Hercynian’ inversion and/or by hot fluid flow along the plane of the ‘Hercynian’ Unconformity. Unfortunately, however, Marshall (pers. com., 2025) had not been given access to the well logs or subsequent re-evaluation of the well and so was unaware of either the presence of a thick dolerite sill in Well 348-3 (Fig. 1) or the fact that the stratigraphy of the well had been reinterpreted as being entirely below the ‘Hercynian’ Unconformity. Therefore, other possible mechanisms should also be considered. These are reworking, ‘contact maturation’ associated with intrusive igneous activity, and advective heating by hot fluids:

### *Deep weathering*

Deep weathering associated with either ‘Hercynian’ inversion or present-day exposure are both highly unlikely, as samples A and B, both >600 ft higher in the section than sample C, are unaffected.

### *Fluid flow along the plane of the ‘Hercynian’ Unconformity*

For the same reason as deep weathering, this mechanism can be discounted.

### *Reworking*

The two darker acritarch populations from sample C both comprise acritarch taxa such as? *Ammonidium* cf. *microcladum*, *Ammonidium* sp., *Arkonia nova*, *Crassianguilata variacornuta*, *Diexallophasismicrodenticulata*, *Neoverhachium carminae* and *Ovinia* cf. *desertica*, all of which are typical of the early Silurian; no taxa were recorded that are restricted stratigraphically to the pre-Silurian. For the anomalously dark acritarchs to have been reworked, an early Silurian source section elsewhere would need to have been deposited, buried to a depth of several km, exhumed, and finally eroded, transported and re-deposited, all during early Silurian time. This timescale is considered so unlikely that reworking can be discounted.

### *‘Contact maturation’*

The quartz dolerite dyke from approximately 300 ft below sample C is ca. 110 ft thick with a Rb-Sr age of 365 ma. Many studies have described maturation aureoles of igneous intrusions but one of the most relevant is that published by Bulguroglu and Milkov (2020) who investigated Permian sedimentary rocks intruded by Jurassic dolerite sills in Namibia and concluded that the maximum thickness of the maturation aureole for sills >13m thick is 135% of sill thickness, equating to ca. 150 ft for the sill in Well 348-3. Therefore, conductive heating by the sill can be eliminated as a cause of the anomalously high PDI values in this well.

### *Advective heating by hot fluids*

Too little data exists for the effects of advective heating to be modelled effectively but by elimination of other potential mechanisms, this appears to be the probable cause of the anomalously high maturity of sample



C. Calculation of the peak temperatures indicated by the two highest modal classes in Fig. 3(c) using the equation of Barker and Pawlewicz (1986) indicates minimum peak temperatures within the approximate ranges of 135–150 and 150–162 °C. A somewhat speculative interpretation of these results would be that rocks around the depth of sample C were affected by two pulses of hot fluids associated with the intrusion and cooling of the dolerite sill in late Devonian time (365 ma). The second, hotter pulse with a minimum peak temperature of 150–162°C may have been of shorter duration than the first pulse (135–150 °C) insufficient for the PDI signature of the latter to be completely overwritten, i.e. not all the acritarchs were darkened further. This interpretation is fully consistent with Marshall's generalised interpretation of the early Palaeozoic succession in the area having been subjected to a short period of very high heat flux in Silurian time.

## Conclusions

Acritarch Palynomorph Darkness Index (PDI) is a viable method for estimating the thermal maturity of pre-late Devonian rocks.

The calibration of Acritarch PDI to Vitrinite Reflectance can be improved by using PDI samples whose maturity has been established using a chitinozoan-based 'Vitrinite Reflectance Equivalent'.

When integrated with relevant geological data, Acritarch PDI is a useful tool for elucidating thermal history.

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## References

- Barker, C.E. and Pawlewicz, M.J., 1986. The correlation of vitrinite reflectance with maximum temperatures in humic organic matter. In: G. Buntebarth and L. Stegena (Editors), *Palaeogeothermics*. Springer, Berlin, 79–93.
- Bulguroglu, M.E. and Milkov, A.V., 2020. Thickness matters: influence of dolerite sills on the thermal maturity of surrounding rocks in a coal bed methane play in Botswana. *Marine and Petroleum Geology* **111** (2020) 219–229.
- Buratti, N., De Luca, R., Garuttii, L.A., Spina, A. and Clayton, G. 2024. Thermal maturity evaluation on mildly artificially oxidized sporomorphs: A comprehensive calibration of palynomorph darkness index (PDI) with vitrinite reflectance. *Marine and Petroleum Geology* **162** (2024) 106672.
- Clayton, G., Goodhue, R., Abdelbagi, S.T. and Vecoli, M., 2017. Correlation of Palynomorph Darkness Index and vitrinite reflectance in a submature Carboniferous well section in northern Saudi Arabia. *Revue de Micropaleontologie* **60**, 411–416. <https://doi.org/10.1016/j.revmic.2017.07.002>.
- Clayton, G., Vecoli, M., Pan Luo, Goodhue, R. and Wellman, C. 2024. Acritarch Palynomorph Darkness Index (PDI) as an indicator of thermal maturity in Silurian sections from Saudi Arabia. *Marine and Petroleum Geology* **169** (2024) 107049.
- Evenick, J.C., 2021. Examining the relationship between Tmax and vitrinite reflectance: an empirical comparison between thermal maturity indicators. *J. Nat. Gas Sci. Eng.* **91** (2021), 103946.
- Goodhue, R. and Clayton, G., 2010. Palynomorph darkness index (PDI) – a new technique for assessing thermal maturity. *Palynology* **34**, 147–156. <https://doi.org/10.1080/01916121003696932>.
- Marshall, J.E.A. 1995. The Silurian of Saudi Arabia: thermal maturity, burial history and geotectonic environment. *Revue of Palaeobotany and Palynology*, **89** (1995), pp. 139–150.
- Spina, A., Vecoli, M., Riboulleau, A., Clayton, G., Cirilli, S., Di Michele, A., Marcogiuseppe, A., Rettori, R., Sassi, P., Servais, T. and Riquier, L., 2018. Application of Palynomorph Darkness Index (PDI) to assess the thermal maturity of palynomorphs: a case study from North Africa. *International Journal of Coal Geology* **188**, 64–78. <https://doi.org/10.1016/j.coal.2018.02.001>.
- Tricker, P.M., Marshall, J.E.A. and Badham, T.J., 1992. Chitinozoan reflectivity: an innovative Lower Palaeozoic maturity indicator. *Marine and Petroleum Geology* **9**, 302–307.